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Monsoon, 2025

B. Tech 5th Semester Examination

STRUCTURAL DESIGN -I

Course Code: BCE23501T

Full Marks – 50

Time – 2 hours

The figure in the margin indicates full marks for the questions.

Part A

Answer ALL questions

Multiple Choice Questions (10 x 1 mark = 10 marks)

1. a) For a given section, if concrete grade is doubled, the cracking moment approximately:
 - (i) Remains same
 - (ii) Increases by $\sqrt{2}$ times
 - (iii) Increases by 2 times
 - (iv) Increases by 4 times
- b) If a rectangular beam section with $b = 300\text{mm}$, $d = 500\text{mm}$, M20, Fe 415, with 4–20 mm diameter bars in tension, then the section is:
 - ☒ (i) Under-reinforced
 - (ii) Over-reinforced
 - (iii) Balanced
 - (iv) Doubly reinforced
- c) The modular ratio (m) for steel and concrete is defined as:
 - (i) $\frac{E_s}{E_c}$
 - (ii) $\frac{E_c}{E_s}$
 - (iii) $1 + \frac{E_s}{E_c}$
 - (iv) $\frac{1 + E_s}{E_c}$
- d) The maximum reinforcement percentage in a column is
 - (i) 2%
 - (ii) 3%
 - (iii) 4%
 - ☒ (iv) 6%
- e) The minimum reinforcement percentage for mild steel in slabs as per IS 456 is
 - (i) 0.10%
 - ☒ (ii) 0.12%
 - (iii) 1%
 - (iv) 1.2%
- f) As per IS 456, a slab is classified as a one-way slab when
 - (i) $L_y/L_x = 2$
 - ☒ (ii) $L_y/L_x > 2$
 - (iii) $L_y/L_x < 2$
 - (iv) $L_x/L_y > 2$
- g) Lateral ties in a column are designed mainly to:
 - (i) Prevent crushing
 - (ii) Prevent buckling of longitudinal bars
 - (iii) Reduce bending
 - (iv) Increase axial capacity
- h) The vertical portion of a step in a staircase is known as _____.
 - ☒ (i) Riser
 - (ii) Tread
 - (iii) Nosing
 - (iv) Flight
- i) The effective span of a stair slab spanning between supports is taken as
 - (i) Clear span only
 - (ii) Center to center distance of supports
 - (iii) Clear span plus effective depth
 - (iv) Clear span plus bearing

j) IS 13920 deals with _____.

(i) Steel welding

(iii) Water-proofing

(ii) Ductile detailing for earthquake resistance

(iv) Ductile detailing for Fire resistance

Part B

Answer ANY THREE questions

(3 x 2 mark = 6 marks) (Max. 100 words each)

2. Explain why minimum reinforcement is necessary in RC beams.

3. Describe the difference between one-way and two-way slabs.

4. Explain the concept of development length.

5. Explain the provisions of splicing of longitudinal reinforcements according to IS 13920.

6. Explain why ductile detailing is essential for reinforced concrete structures in seismic zones as per IS 13920.

7. Differentiate between Working Stress method and Limit State method.

Part C

Answer ANY TWO questions

(2 x 4 mark = 8 marks) (Max. 100 words each)

8. Calculate the moment of resistance of a singly reinforced beam of 250 mm x 450 mm size having 3-20 mm diameter bar as tensile reinforcement. The beam has M25 concrete grade and Fe 415 steel grade.

9. A reinforced concrete beam has $b = 230$ mm and $d = 450$ mm and is made of M20 concrete & Fe415. If the factored shear force at the support is 160 kN, Solve the nominal shear stress and check whether the beam requires shear reinforcement as per IS 456 if percentage of provided main reinforcement is 0.2%.

10. Apply IS 456 and compute the effective depth required for a simply supported slab spanning 1.5 m x 4.1 m for a total design load of 10 kN/m². Adopt any necessary data.

11. Compute and choose the development length of a 16 mm Fe 415 bar in M30 concrete.

Part D

Answer ANY TWO questions

(2 x 8 mark = 16 marks) (Max. 200 words each including subdivisions)

12. A simply supported reinforced concrete beam has a rectangular cross-section 300 mm x 600 mm and is reinforced with 4-25 mm $\varnothing$ tension bars. Take effective cover = 50 mm, M20 concrete and Fe 415 steel. Calculate the cracking moment of resistance M_{cr} of the beam using the Working Stress Method. Evaluate and justify whether this beam is likely to crack under a mid-span bending moment of 120 kNm.

13. Two alternative materials are proposed for a column of effective length 3 m and size 300 mm x 300 mm with minimum 4 numbers of bars: (i) M30 concrete with Fe 415 steel and (ii) M40 concrete with Fe 500 steel. Evaluate which combination provides better safety margin for a 500 kN factored axial load. Adopt any necessary data.

14. A column of 300 mm x 300 mm is having a square footing of 2.1 m x 2.1 m which is distributing a net soil pressure of 0.325 N/mm^2 . The footing is having an effective depth of 344 mm and percentage of steel as 1%. Assess the safety of the footing in one-way and two-way shear. Take M20 concrete and Fe 415 steel.

Part E

Answer ANY ONE questions

(1 x 10 mark = 10 marks) (Max. 300 words each including subdivisions)

15. Estimate the moment of resistance of a T-beam, having width of flange = 900 mm, thickness of flange = 150 mm, width of web = 300 mm, effective depth = 650 mm, and area of tensile reinforcement = 4000 mm^2 . Use M20 concrete and Fe 415 steel.

16. Design a short axially loaded reinforced concrete column of diameter 500 mm to carry a factored load of 1200 kN. Use M20 concrete and Fe 415 steel. Provide the reinforcement details and check for spacing of ties. *Leff = 3.1 m*

17. Design the flight of a dog-legged reinforced concrete staircase for a floor-to-floor height of 3.3 m. The landing beam width is 300 mm, and the staircase is designed to carry a live load of 6 kN/m^2 . Assume any other data required for the design and provide necessary diagrams.